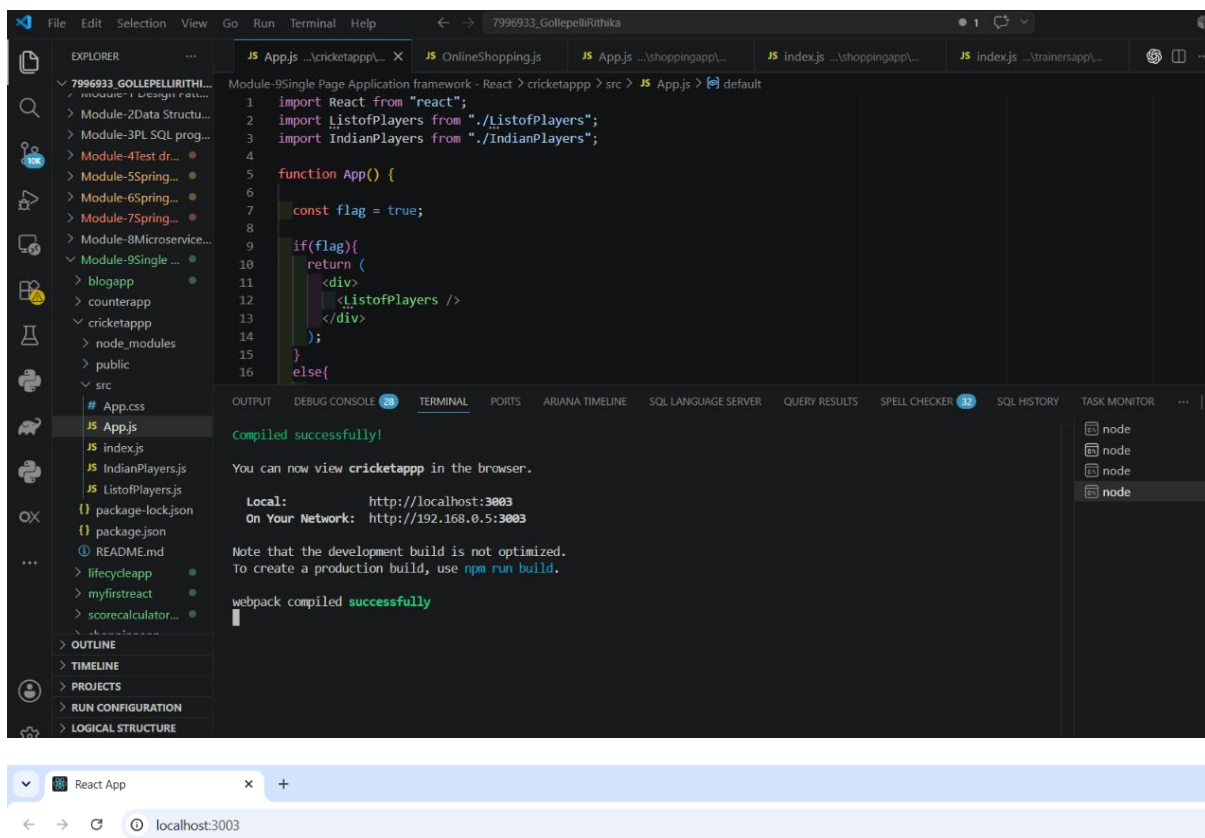


# Output9cricketapp.ReactJS-HOL



## List of Players

- Virat - 90
- Rohit - 75
- Gill - 65
- Rahul - 85
- Pant - 55
- Hardik - 80
- Jadeja - 68
- Ashwin - 60
- Bumrah - 95
- Shami - 72
- Siraj - 50

## Players with Score Below 70

- Gill - 65
- Pant - 55
- Jadeja - 68
- Ashwin - 60
- Siraj - 50

## Objectives

### 1. List the Features of ES6

ECMAScript 6 (**E**uropean **C**omputer **M**anufacturers **A**ssociation) (ES6), also known as ECMAScript 2015, introduced many modern features that make JavaScript easier to write, read, and maintain. These features reduce code complexity and improve application performance.

## Key Features of ES6

- let and const for block-scoped variables.
- Arrow functions (=>) for shorter function syntax.
- Classes for object-oriented programming.
- Template literals using backticks (`).
- Destructuring assignment for arrays and objects.
- Default function parameters.
- Rest (...) and Spread (...) operators.
- Modules (import and export).
- Promises for asynchronous programming.
- Enhanced object literals.
- Map and Set collections.

## 2. Explain JavaScript let

The let keyword is used to declare variables with block scope, meaning the variable is accessible only within the block ({} ) where it is declared. Unlike var, let prevents accidental redeclaration in the same scope and helps write safer, more predictable code.

## 3. Identify the Differences Between var and let

| <b>var</b>                                         | <b>let</b>                                                          |
|----------------------------------------------------|---------------------------------------------------------------------|
| Function scoped.                                   | Block scoped.                                                       |
| Can be redeclared in the same scope.               | Cannot be redeclared in the same scope.                             |
| Hoisted and initialized with undefined.            | Hoisted but cannot be used before declaration (Temporal Dead Zone). |
| May cause unexpected behavior in loops and blocks. | Safer and more predictable for modern JavaScript development.       |
| Used in older JavaScript versions.                 | Recommended in ES6 and later.                                       |

## 4. Explain JavaScript const

The const keyword is used to declare variables whose references cannot be reassigned after initialization. It is also block scoped like let. Although a const variable cannot be reassigned, the contents of objects and arrays declared with const can still be modified.

## 5. Explain ES6 Class Fundamentals

ES6 introduced classes, providing a cleaner and more structured way to create objects and implement object-oriented programming. A class contains properties (data) and methods (functions) that define an object's behavior.

## 6. Explain ES6 Class Inheritance

Inheritance allows one class to acquire the properties and methods of another class using the `extends` keyword. This promotes code reuse and simplifies application development.

## 7. Define ES6 Arrow Functions

**Arrow functions** (`=>`) provide a shorter and more readable syntax for writing functions. They automatically inherit the `this` value from their surrounding scope, making them especially useful in callbacks and event handlers.

### Benefits

- Shorter syntax.
- Easier to read.
- No need to write the `function` keyword.
- Automatically binds the surrounding `this`.

## 8. Identify Set() and Map()

### Set

A **Set** is a collection that stores **only unique values**. Duplicate values are automatically removed.

### Example

```
const numbers = new Set([10, 20, 20, 30]);  
console.log(numbers);
```

### Output

Set(3) {10, 20, 30}

### Features

- Stores only unique values.
- Maintains insertion order.
- Supports methods like `add()`, `delete()`, and `has()`.

### Map

A **Map** is a collection of **key-value pairs**, where keys can be of any data type.